

Ten Layers at the Rank-Growing Determinant Frontier

RH-222–RH-230 Review and the Uniform Complement-Ideal Wall

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Abstract

RH-221 ends with a finite gauge-complete quartet and an exact requirement: the next determinant candidate must add genuinely new roots. This review collects nine subsequent layers.

RH-222–RH-223 construct shell-complete conjugate resonance clouds whose ranks grow from four to 34–35 across sixteen scales and two channels. Naive fixed-cardinality prefixes split a conjugate pair at 23 of 32 endpoints, whereas minimal shell completion is stable under four candidate margins.

RH-224 proves that one global center and RMS radius make every empirical cloud measure uniformly tight. RH-225 then proves the complementary obstruction: a tight probability cloud with rank tending to infinity cannot itself be a locally finite unweighted zero divisor. The correct Fredholm zeros are reciprocal resonances. RH-226 restores that exact dictionary and reconnects the finite atlas to RH-7’s rigorous fixed-positive-noise Hilbert–Schmidt determinant.

The small-noise problem remains singular. Reciprocal zero counts fail the frozen local-count gate. Twelve to fourteen resolved omitted roots per endpoint have a successful unit-disk logarithmic tail bound, but the only whole-matrix Frobenius certificate grows from 5.15 to 169.27, approximately as $\sigma^{-1.03}$. All sixteen dual-channel determinant comparisons pass a 0.02 gate, while adjacent-scale differences do not contract.

The new coordinate is

`rank_growing_reciprocal_cloud_open_uniform_complement_ideal_limit.`

The aggregate ledger contains 9,870 finite cases and zero identity failures. Gate A remains open; Gates B–E, Hilbert–Polya, zeta-divisor equality, and RH are untouched.

1 Coordinate inherited from RH-221

RH-212–RH-220 classify the fixed outer quartet exactly and then prove that a fixed quartic cannot supply a growing locally finite spectral count [4, 5]. RH-221 leaves the coordinate

`finite_gauge_complete_shape_flow _open_rank_growing_divisor.` (1)

The recommended next route has five logically distinct tasks:

1. build a rank-growing physical root atlas;
2. apply one global gauge and prove empirical tightness;
3. determine the correct divisor variable;
4. control omitted regularized-product factors;

5. obtain a locally uniform determinant family.

RH-222–RH-230 complete the first three at the finite or exact level, partially complete the fourth, and expose the wall in the fifth.

2 Layer-by-layer ledger

paper	principal conclusion	status
RH-222	shell-complete clouds grow from rank 4 to 34–35	finite construction
RH-223	naive cut splits pairs; completed cut is margin stable	exact + finite
RH-224	global normalized empirical measures are uniformly tight	exact theorem
RH-225	tight rank growth cannot be a direct locally finite divisor	exact obstruction
RH-226	determinant zeros are reciprocal resonances; fixed-noise $\det_2$ reconnected	exact + inherited
RH-227	reciprocal local counts do not pass frozen tail stability	finite negative
RH-228	resolved omitted shells have a quantitative unit-disk tail	exact + finite positive
RH-229	whole-matrix Frobenius tail certificate is unusably large	finite negative
RH-230	dual channels agree; adjacent scales do not contract	finite mixed

The central conceptual correction is that empirical roots and Fredholm zeros are not the same objects. The former are useful for global geometry; the latter determine holomorphic local finiteness.

3 RH-222–RH-223: a symmetry-correct rank-growing atlas

At each frozen scale, RH-222 resolves the folded Gaussian fine operator and its Haar compression, removes Perron and negative parity resonances, and Hardy-scales the bulk by 0.85 [6]. The target ladder is

$$k_\ell = 4 + 2\ell, \quad 0 \leq \ell \leq 15.$$

Real eigenvalues form singleton shells and nonreal eigenvalues form conjugate pair shells. Whole shells are included until the target is reached.

Actual ranks increase strictly on both channels. They range from four to 35 on the left and four to 34 on the right. The selected conjugacy error is zero at archived precision, overshoot is at most one, and the smallest resolved post-cloud gap is 7.3991×10^{-5} .

RH-223 proves minimality of this completion [7]. A naive exact target prefix splits a pair in 23 of 32 cases. Exactly those 23 cases require one-root overshoot. Prefix margins 4, 8, 12, 14 all recover the reference multiset with zero matching error.

The selection is canonical only relative to the resolved radial ordering. No all-level rank law or interval-certified Riesz separation is obtained.

4 RH-224: probability compactness

For a cloud Λ of rank n , define

$$m_\Lambda = \frac{1}{n} \sum_{\lambda \in \Lambda} \lambda, \quad s_\Lambda^2 = \frac{1}{n} \sum_{\lambda \in \Lambda} |\lambda - m_\Lambda|^2$$

and

$$\mu_\Lambda = \frac{1}{n} \sum_{\lambda \in \Lambda} \delta_{(\lambda - m_\Lambda)/s_\Lambda}.$$

Then

$$\int z d\mu_\Lambda = 0, \quad \int |z|^2 d\mu_\Lambda = 1. \quad (2)$$

Markov's inequality gives

$$\mu_\Lambda(|z| > R) \leq R^{-2}. \quad (3)$$

Therefore every globally normalized cloud family is uniformly tight and has weakly convergent subsequences by Prokhorov compactness [1, 8].

The numerical atlas has maximum normalized modulus 1.83188 and second-moment error below 6.7×10^{-16} . Tightness is exact, while uniqueness of the weak limit remains open.

5 RH-225: why tight roots cannot be direct zeros

Let

$$\nu_n = \sum_{\lambda \in \Lambda_n} \delta_\lambda, \quad \mu_n = \frac{1}{d_n} \nu_n, \quad d_n \rightarrow \infty.$$

Uniform tightness puts at least half of each probability measure in one compact set K . Hence

$$\nu_n(K) \geq d_n/2 \rightarrow \infty.$$

The integer divisors cannot converge vaguely to a locally finite divisor. They also cannot be complete zero divisors of a holomorphic family converging locally uniformly to a nonzero limit [2, 9].

This exact result rules out globally normalized resonances as determinant zeros. In the finite atlas the direct compact counts grow by 31 and 30 roots. All raw selected resonances also lie in the unit disk, so the same finite failure is visible before global normalization.

The result does not apply after inversion. If compact-operator resonances λ_j tend to zero, reciprocal points $1/\lambda_j$ escape to infinity.

6 RH-226: the Fredholm variable

For

$$p_\Lambda(w) = \prod_{\lambda \in \Lambda} (w - \lambda),$$

RH-226 restores the identity

$$\prod_{\lambda \in \Lambda} (1 - z\lambda) = z^{|\Lambda|} p_\Lambda(1/z). \quad (4)$$

The second-regularized factor

$$P_{\Lambda,2}(z) = \prod_{\lambda \in \Lambda} (1 - z\lambda) e^{z\lambda} \quad (5)$$

has the same zeros, namely $z = 1/\lambda$ [3, 10].

The physical reciprocal moduli range from 1.15129 to 13.54281. Equation (4) has maximum grid error 9.16×10^{-15} .

This does not construct a new fixed-noise determinant. RH-7 already proves that the centered Gaussian transfer operator is Hilbert–Schmidt and that its $\det_2$ is entire with reciprocal resonance zeros [15]. The new question is uniformity as noise decreases.

7 RH-227: local zero counts do not stabilize yet

If $f_\sigma \rightarrow f \neq 0$ locally uniformly, Rouché's theorem forces eventual zero-count stability on every contour avoiding zeros of f . RH-227 tests the selected reciprocal clouds on

$$R = 1.2, 1.5, 2, 3, 5$$

[11].

Only one of ten side-radius sequences is constant over the last four scales. The largest first-to-last count growth is 18, and the minimum finite contour clearance is 1.3338×10^{-4} . Radius-two counts grow from two to ten on both channels.

Finite nonstability does not disprove eventual stabilization. It shows that a raw small-noise limit is not supported by the current range and that near-unit factors cannot be ignored.

8 RH-228: resolved shells are controllable

For $|w| < 1$,

$$|\log(1 - w) + w| \leq \frac{|w|^2}{2(1 - |w|)}.$$

If T is an omitted resonance multiset and $q = R \max_{\lambda \in T} |\lambda| < 1$, then

$$\sup_{|z| \leq R} |\log P_{T,2}(z)| \leq \frac{R^2}{2(1 - q)} \sum_{\lambda \in T} |\lambda|^2. \quad (6)$$

Every endpoint has 12–14 complete resolved roots beyond its selected cloud. On $|z| \leq 1$, the worst q is 0.29818, the worst bound is 0.16804, and the worst observed grid tail is 0.07532 [12]. This is a genuine positive omitted-factor result, but only inside the resolved candidate window.

9 RH-229: the full complement wall

Schur triangularization gives

$$\sum_j |\lambda_j(A)|^2 \leq \|A\|_{\mathbb{F}}^2.$$

After subtracting selected Perron, parity, and cloud eigenvalue masses, the remainder bounds the squared mass of all omitted finite-matrix eigenvalues. Combining it with (6) produces a whole-matrix certificate.

The certificate fails 0/32 unit-disk gates. It ranges from

$$5.15135 \quad \text{to} \quad 169.27116$$

and grows descriptively as $\sigma^{-1.025}$ and $\sigma^{-1.033}$ [13]. It can exceed the resolved-shell bound by a factor 5563.49.

This is a certificate obstruction, not a determinant obstruction. The Frobenius norm sees nonnormal singular mass represented by off-diagonal Schur entries. A reducing complement or two-step trace-class estimate may be much smaller.

10 RH-230: coherent channels, noncontracting scales

Selected factor logarithms are compared on 256 points in $|z| \leq 1$:

$$L_{\Lambda}(z) = \sum_{\lambda \in \Lambda} [\log(1 - z\lambda) + z\lambda].$$

All sixteen same-scale channel differences are below 0.02, with maximum 0.017855. Thirty adjacent-scale differences range from 0.042131 to 0.183660, and neither channel decreases strictly over the final four transitions [14].

The determinant representation therefore preserves finite dual-channel coherence but supplies no Cauchy evidence across scale. This mixed result supports one shared renormalization, not convergence of the selected factor alone.

11 Aggregate reproducibility ledger

The batch counts:

1. endpoint, channel, and candidate-margin cases;
2. exact gauge and tail checks;
3. 3,072 reciprocal-polynomial evaluations;
4. 6,144 resolved-tail grid evaluations;
5. local-count, Frobenius-budget, and determinant-comparison cases.

The total is 9,870 finite ledger items. Six independent identity families have zero failures:

conjugacy, margin matching, RMS moments,
probability tails, reciprocal identity, regularized tail bound.

Finite test success does not alter theorem status. In particular grid comparisons are not interval-validated suprema and Arnoldi roots are not certified Riesz data.

12 Revised Gate-A route

The coordinate after this batch is

$$\boxed{\begin{array}{l} \text{rank_growing_reciprocal_cloud} \\ \text{_open_uniform_complement_ideal_limit} \end{array}}. \quad (7)$$

The next target is not another empirical normalization. It is a physical moving-cloud relative determinant:

1. construct a Riesz or reducing projection for the near-unit cloud;
2. factor its finite $\det_2$ contribution without losing conjugate and dual-channel coherence;
3. bound the complementary block uniformly in Hilbert–Schmidt norm, or bound its square uniformly in trace norm, on a nonshrinking z -disk;
4. prove local reciprocal-zero count stabilization for the complement;

5. take a locally uniform relative-determinant subsequential limit and show it is nonzero.

RH-80 supplies an abstract moving-cloud relative determinant template [16]. The new work must connect that template to the operator-derived RH-222 shells and overcome the nonnormal Frobenius barrier. This is the recommended RH-232 target.

13 Macro-gate boundary

Gate A is not closed: there is no uniform small-noise relative determinant or dynamical limiting realization. Consequently:

- Gate B, a time-oriented unitary or scattering completion, is untouched;
- Gate C, a self-adjoint generator and internally derived $T \log T$ count, is untouched;
- Gate D, a prime-power/von Mangoldt trace identity, is untouched;
- Gate E, equality with the completed zeta divisor, is untouched.

No Markov resonance or reciprocal zero is identified with a Riemann zero. No Hilbert–Polya operator or Riemann-hypothesis implication is asserted. The batch advances the internal determinant architecture and identifies its next precise wall.

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